

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

Total Marks: 40

QUESTIONS: 4

Annexure A

CASE STUDY

Code of Conduct:

I, M. C. Kulkarni 192565067

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



23+23+17+17+8
(28)
bf

ten Rules.

Failed login attempts AND unusual IP address $\rightarrow$ possible brute force attack.
unusual IP address AND High data transfer $\rightarrow$ possible Data Exfiltration

a) propositional logic Representation

F = Failed login Attempts

U = unusual IP address

O = odd login time

H = High data transfer

B = possible Brute force attack

D = possible Data Exfiltration.

Rule 1: $F \wedge U \rightarrow B$

Rule 2: $F \wedge O \rightarrow B$

Rule 3: $U \wedge H \rightarrow D$

facts

F = True

U = True

H = True

Goal
possible Data exfiltration

Rule 3: $U \wedge H \rightarrow D$

unusual IP address
True

High data Transfer
True

Goal proven.

possible Data Exfiltration = True

Case Study - Airport priority landing system

Rule:

Aircraft (x) is Emergency Declared (x) $\rightarrow$ priority clearance

Aircraft (flight z)

Emergency Declared (flight z).

$\phi = \{x \mid \text{Flight } z\}$

priority clearance (flight z)

Runway Reserved (flight z)

b) Resolution:

Goal:

Aircraft (flight x) $\vee$ Delay (flight x) $\vee$ Queued After (flight x, flight z) $\vee$ Runway Reserved (flight z) $\vee$ Delay (flight x)

Aircraft (flight x)
Queued After (flight x, flight z)
Runway Reserved (flight z)

Delay (flight x)

c) Two emergency Aircraft

- * Emergency severity
- * priority comparison
- * Runway allocation
- * Conflict resolution.

Equipment

M = Machine overhauling
N = Maintenance needed
H = High priority line
S = Schedule modification
P = Production

Industrial equipment Maintenance

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Let:

M = Machine overheating

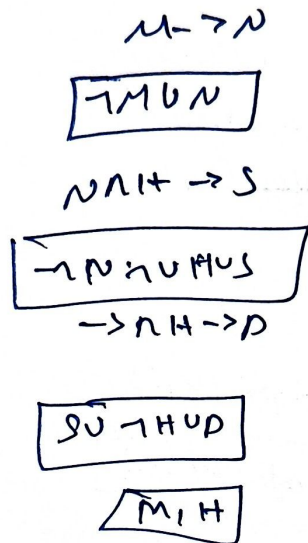
N = Maintenance Needed

H = High priority line

S = Schedule immediate repairs

P = Production Risk

a) NF Conversion



b)

Resolution.

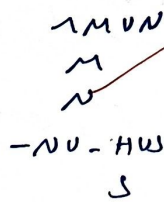
from:

and:

derive:

from:

and M, H, derive



therefore:

c) If Machine crushing is removed.
without M: $M \rightarrow N$.

Therefore, S cannot be derived.

Also, production Risk does not follow because $\rightarrow$ is not explicitly known.

Production Risk cannot be proved

a) Belief state:

Vibration [2,1]
Smoke [1,1]
Beacon [2,2]

Mine Adjacent [2,1]
Fire Adjacent [1,1],
Safe zone marked [2,2]

b) forward chaining
from:

derive:

Vibration [2,1]
Mine Adjacent [2,1]
Smoke [1,1]

derive:

Fire Adjacent [1,1]

from:

Beacon [2,2]

Safe zone marked [2,2]

path Safety

path:

[1,1] fire warning
[2,1] Mine warning
[2,2] Safe zone marked

[1,1] $\rightarrow$ [2,1] $\rightarrow$ [2,2]

Therefore:

The path is not guaranteed to be safe